

Water Contamination from Industrial Livestock Production

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Abstract

Over the past century, U.S. livestock production has shifted from many small farms to fewer, increasingly larger operations. Today, most animals are raised in confined Animal Feeding Operations (AFOs), where manure is stored on-site and eventually applied to fields. This concentration of waste has long raised concerns about pollution, yet credible causal evidence on its impacts remains scarce. I assemble new panel data linking AFOs in Iowa and North Carolina to downstream water monitoring records and delineate station-specific drainage basins to capture the hydrological structure of pollutant transport. The analysis exploits two complementary research designs. In both states, I implement an event-study approach leveraging extreme precipitation shocks at the location of the confinement sites. In Iowa, I additionally implement a difference-in-differences design exploiting spatio-temporal variation in the number and size of AFOs. Across both settings, I find that AFOs significantly degrade surface water quality. Both chronic and acute responses are largely concave; chronic effects scale with herd size, and pollution loads respond to total livestock burden rather than its distribution across larger or smaller operations. With global livestock production projected to grow and intensify further, and with regulatory changes in some regions moving toward weaker environmental oversight, these findings provide timely evidence on the pollution risks of large-scale operations.

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1 Introduction

Over the last century, U.S. livestock production has undergone a structural transformation from many small farms to fewer and increasingly larger operations (MacDonald & McBride, 2009). The total number of animal farms in the U.S. declined from around 1,517,000 in 1982 to 868,000 in 2022, while sales of U.S. animal products increased from 179 to 262 billion US dollars (in dollars of 2022) (USDA NASS, 1984, 2024). In the hog production sector, in the largest producing state—Iowa—the number of farms dropped by 89% (45,800 to 5,300), while the annual number of hogs sold rose from 24 to 60 millions in total across farms, and from 500 to 9,700 on average per farm. These trends were achieved by the gradual transition to large and intensive confined operations, called Animal Feeding Operations (AFOs), which now represent the overwhelming majority of animal farms across states (MacDonald & McBride, 2009; McBride & Key, 2013).

In these industrial operations, animals are mostly kept in confinement buildings, and produce manure which is temporarily stored on site. While the animals themselves are subject to FDA regulation, there is limited oversight of the pollutants emitted by these operations. Yet they represent a potentially large threat to environmental and public health as well as rural quality of life. Indeed, a single farm can produce more manure than all the sanitary waste generated by a city of 1.5 million residents (U.S. GAO, 2008), yet unlike city sewage, animal waste is untreated (Hribar, 2010). This waste releases harmful gases as it decomposes, such as ammonia and hydrogen sulfide, and contains various pollutants, including excess nutrients and harmful pathogens (U.S. EPA, 2013). In all, U.S. AFOs produce over 1.2 billion tons of waste annually, reaching levels potentially severe enough to have triggered a 2005 Congressional hearing titled “Superfund Laws and Animal Agriculture” (U.S. House of Representatives, 2005). In addition, residues from the frequent administration of antimicrobials—often at low doses and over extended periods—if disseminated, may contribute to the spread of antimicrobial resistance and increase the risk of resistant infections in human populations. The confinement buildings themselves also emanate particulate matter. The environmental and public health concern is that through airborne and waterborne transmission, these contaminants do not stay enclosed in feeding operations but instead put the surrounding communities at risk of environmental exposure (Mallin, 2000). The water exposure pathway is of particular concern for operations that store manure in liquid form, which is the case for the vast majority of swine and dairy operations.

Numerous studies document environmental exposures and their impacts on both ecosystems and human health. However, these impacts have either not been assessed at large scale or not been robustly quantified. Afflictions associated with proximity to AFO practices include respiratory problems, digestive disorders, impaired mental health, low birth weight, and infant mortality, and are thought to be mediated by air and water transmission. Some causal evidence has been provided for the airborne exposure pathway; outcomes analyzed include air pollution measures (Sneeringer, 2010), birth outcomes (Sneeringer, 2009) and property values (Isakson & Ecker, 2008; Kim & Goldsmith, 2009; Lawley, 2021; Palmquist et al., 1997). The water exposure pathway, by contrast, has not yet been robustly assessed for causal impacts, despite concerns about both surface water and groundwater pollution. Raff and Meyer (2021) analyzes impacts on surface water nutrient concentrations of AFOs,

focusing on the largest dairy operations in the state of Wisconsin. [Wing and Wolf \(2000\)](#) relies on a matching identification strategy and finds increased occurrences of gastrointestinal problems with proximity to swine operations, suggesting contamination through water exposure, but is limited by a small sample size (155 survey respondents).

Understanding the extent of the impact of environmental exposure to the pollutants from such facilities is ever more important, for multiple reasons. First, global animal production is projected to increase in the short- to medium-term, and in the U.S. as well as abroad, much of this growth seems planned to occur through the expansion of such industrial operations. World meat production is expected to rise by 13% by 2030, with half of this increase corresponding to chicken production and a third to pigmeat ([OECD & FAO, 2021](#)). In the U.S., by 2031, all types of animal production are expected to increase substantially, notably milk by 9.1%, pork by 7.9%, and chicken and egg by around 14% ([USDA, 2022](#)). Two important features of the projected increases in swine production are worth noting: (i) 66% of the global increase would come from Asian countries, particularly China and Vietnam, where the imported model of industrial swine operations is supported by development banks;¹ (ii) the 8% growth in the Americas should be driven notably by “further intensification of production systems”. International organizations also promote the system’s “sustainable intensification” by its “sustainable industrialization” ([FAO, 2018](#)).

Second, in many countries, including the U.S. (as detailed in the next section), regulatory frameworks governing large-scale animal operations commonly rely on size thresholds to trigger meaningful environmental oversight. Following an initial 2017 petition for rule-making filed by environmental interest groups, the U.S. Environmental Protection Agency (U.S. EPA) initiated in 2023 an evaluation of whether to revise its regulations or pursue non-regulatory initiatives. Most recently, in France, similar thresholds were raised substantially—from 40,000 to 85,000 heads of poultry, and from 2,000 to 3,000 swine—while beef operations were exempted, and the process of public consultation for establishing new operations was simplified.² These regulatory revisions are unfolding in a context where robust causal evidence on their impacts remains scarce, yet the accumulation of suggestive evidence points to potentially serious threats to both environmental and human health.

Simultaneously, nuisance lawsuits against AFOs have been on the rise across the U.S. ([Hines, 2018](#); [Sellers, 2017](#); [Smart, 2016](#)), as residents of nearby rural communities claim persistent harms to their quality of life and property values. In response, a second generation of state “right-to-farm” laws has emerged, designed to shield agricultural producers from such litigation.³ These laws often expand protections by exempting from nuisance claims activities

¹Development banks directly invest in the expansion of industrial swine production in Asia. Most recently in February 2022, a US\$52 million investment from the World Bank’s private sector arm, the International Finance Corporation (IFC), was approved to finance the expansion of one of the main vertically integrated pork producer, meat processor, feed mill operator, and veterinary health products manufacturer in Vietnam (<https://disclosures.ifc.org/project-detail/ESRS/45292/mavin>). The IFC further funds industrial animal agriculture in other parts of the world both directly, such as through the expansion of industrial broiler chicken operations in Uganda (<https://disclosures.ifc.org/project-detail/SII/44775/yokuku-mezz>), and indirectly, through proposed loans to support massive monoculture of crops used as feed for the industrial livestock sector, such as soy and corn in Brazil (<https://disclosures.ifc.org/project-detail/ESRS/44281/ldc-brasil>).

²These provisions were initially introduced by decree in June 2024, then enshrined in law in 2025.

³Starting in the 1970s, when suburban encroachment on rural areas was bringing new residents close

defined as “normal agricultural practices” or “minor changes” in operation. Yet this raises a critical question: should, e.g., the conversion of cropland into an intensive animal feeding operation, or the substantial expansion of an existing AFO, be considered a “minor change” within the scope of normal farming? The very large damages awarded by courts—upwards of the millions of dollars per affected household (Gore, 2020; Pollard, 2020; Anderson v. Murphy-Brown LLC, 2018)—which threaten the economic viability of these operations and point to the magnitude of the externalities they seem to generate, highlight how much turns on where the legal boundary of “normal agriculture” is drawn. At the same time, rigorous causal evidence quantifying the pollution harms of AFOs remains limited, leaving policy and legal debates to proceed in a context of suggestive but not yet definitive scientific findings.

Understanding the causal impact of AFOs on human health through the environmental exposure pathway is therefore both timely and of high policy relevance.

While substantial evidence links AFOs to adverse local outcomes through water-borne pollution, there is limited causal evidence on the magnitude and structure of these externalities. This paper provides causal evidence on surface water contamination using exact upstream-downstream hydrological relationships, thereby establishing a key environmental exposure channel from intensive livestock production. I study Iowa and North Carolina, two states with high concentrations of animal feeding operations and contrasting regulatory environments. Specifically, I test whether upstream AFO activity measurably degrades downstream surface water quality and estimate the magnitude of these effects across multiple pollution indicators of relevance for environmental and public health.

The empirical strategy draws on two complementary sources of quasi-experimental variation. Where spatial and temporal variation in upstream livestock intensity is sufficiently rich to be informative—namely in Iowa—I exploit this variation using a difference-in-differences framework to identify chronic pollution effects. In both Iowa and North Carolina, I additionally exploit exogenous extreme precipitation events at upstream facility locations to identify acute pollution responses driven by shocks to pollutant transport. By leveraging detailed geospatial data on AFO locations and hydrologically defined drainage areas, the analysis captures both persistent and episodic dimensions of surface water contamination from animal agriculture.

to industrial animal operations, the threat of nuisance lawsuits with costs that would push farmers out of business or prevent them from investing in their farm led states to legislate the “right to farm” (RTF). The first generation of RTF legislation prohibited nuisance lawsuits by new neighboring residents against farmers when “normal farming operations” preexisted their move. An individual could file a claim only if a new operation was developed after their arrival, or if an existing farm became industrialized (with a significant change on the farm, e.g., in size, hours of operation or technology used). Since the 2000s and especially in the last few years, a second generation of RTF laws has emerged to expand the right to farm. Some states are expanding the scope of their existing legislation, i.e., reducing the circumstances under which farms can be held liable for nuisance — e.g., Indiana in 2005, Florida in 2021 — others are enshrining this right in their constitution — e.g., Missouri in 2014; West Virginia introduced two RTF constitutional amendments in 2021 which failed.

2 Background

Regulation

While the animals themselves are subject to FDA regulation, federal oversight of pollution from AFOs is limited (U.S. GAO, 2008). Air emissions remain unregulated,⁴ and only large or so-called concentrated AFOs, or “CAFOs”, which hold at least 1,000 animal units,⁵ are considered point sources under the Clean Water Act, and are thus subject to the National Pollutant Discharge Elimination System (NPDES) permit regulations. Other provisions, such as Section 404 permitting, also exempt AFOs under the rubric of “normal farming” activities.

The enforcement of the NPDES program is delegated by the U.S. EPA to state agencies in most states.⁶ An NPDES permit is required for a facility to open or expand, but notably the permit covers only the waste management system rather than the operation itself—in effect considering CAFOs as manure-production facilities in the context of environmental oversight. Indeed, the disposal of that manure is the primary environmental concern.

Two broad types of systems are distinguished: dry-manure facilities (mostly poultry) and wet-manure facilities (swine and cattle). The latter pose the greatest risk to water quality. In swine operations in particular, manure is typically liquefied and stored in open pits or “lagoons” on-site before being extracted and applied to sprayfields. Permits require Nutrient Management Plans (NMPs), which must demonstrate that the permit holder has sufficient land available with the right soil composition for nutrients to be applied at “agronomically appropriate” rates under effluent limitation guidelines. However, NMPs regulate only nutrient application and do not address antibiotics, pathogens, or other contaminants—in contrast with the regulation of sewage sludge, for which the EPA has established limits on pathogens and heavy metals.

Most states are authorized to administer the base NPDES program and may impose additional requirements so long as they are at least as stringent as federal standards. In practice, most states have developed AFO-specific programs with their own provisions. In Iowa,

⁴A priori, any facility that emits hazardous air pollutants in quantities such that it would constitute a “major source” would trigger Clean Air Act “Title V” operating permit requirements or reporting requirements under the Comprehensive Environmental Response, Compensation, and Liability Act (CERCLA) and the Emergency Planning and Community Right-to-Know Act (EPCRA). However, such pollutants generated by AFOs remain unregulated, for two reasons: (1) The EPA has not finalized the development of reliable emission estimating methodologies to determine whether AFOs are subject to these requirements (U.S. EPA OIG, 2017); (2) The EPA, countering the 2017 court ruling of *Waterkeeper Alliance et al. v. EPA*, published final rules exempting farms from reporting air emissions from animal waste under CERCLA (August 2018) and EPCRA (June 2019).

⁵The U.S. EPA defines animal units in the regulations that govern NPDES permits [40 C.F.R. § 122]. 1,000 animal units are equivalent to the following animal quantities: 700 mature dairy cows, 1,000 veal calves or beef cattle, 2,500 swine above 55 lbs, 10,000 swine under 55 lbs, 30,000 laying hens or broilers if the poultry AFO uses a liquid manure handling system, and 125,000 chickens or 82,000 laying hens if the poultry AFO uses other than a liquid manure handling system. The term “animal unit” was removed in 2003, but the same animal-equivalent quantities apply.

⁶In Iowa, the Department of Natural Resources is responsible for implementing the NPDES program, and notably issuing permits directly to the discharging facilities. In North Carolina, the Division of Water Resources within the Department of Environmental Quality holds this mandate.

all confinement feeding operations above 1,000 animal units (about 2,500 finishing hogs) require both construction permits and manure management plans, in addition to NPDES coverage. North Carolina, by contrast, has implemented a comprehensive water-quality permitting program that extends beyond federal requirements: since 1993, all AFOs using liquid waste management systems must obtain a permit once they exceed 100 cattle, 75 horses, 250 swine, 1,000 sheep, or 30,000 birds.

Water pollution from manure

The infrastructure and the management practices required under the NMPs do not fully prevent spillage or runoff risks. Open-air lagoons, where most swine facilities store liquid manure, are prone to leaks, breaks, or overflows, and manure spread on saturated soils can leach from the surface (Armstrong et al., 2010; Mallin, 2000; Simpkins et al., 2002).

The sheer volume of waste therefore raises persistent concerns, and there is suggestive evidence of both chronic contamination of waterways—through lagoon leakage and spray-field runoff—and acute spikes during heavy precipitation, as lagoons overflow and soils become saturated. Local analyses downstream of sprayfields show that manure is not fully absorbed, with substantially elevated contaminant levels in both groundwater and surface waters (Harden, 2015; Karr et al., 2001). Through both pathways—lagoons and sprayfields—pollutants reach water bodies, and risks are amplified after heavy rains.⁷ The contaminants of concern include excess nutrients—nitrogen (notably in the forms of nitrate and ammonia) and phosphorus—as well as heavy metals and pathogens. At high concentrations these create both environmental and human health hazards. Coliform bacteria have been linked to gastroenteritis outbreaks; elevated nitrate intake can cause infant methemoglobinemia (“blue baby syndrome”). Nutrient over-enrichment can also have large consequences on ecosystems, as it drives eutrophication, leading to algal blooms, subsequent oxygen depletion, and finally hypoxic “dead zones,” and, in extreme cases, fish die-offs.

Evidence of environmental exposure to AFO pollutants and consequences

A large literature documents positive relationships between proximity to AFO locations and practices and environmental pollution, particularly excess nutrients and pathogens—see for example (Heaney et al., 2015; Mallin, 2000; Messier et al., 2014; Wing et al., 2008). However, most studies rely on limited site samples or document correlations, without establishing causal mechanisms. Raff and Meyer (2021) is a notable exception, providing causal evidence that permitted CAFO dairy production increases surface water concentrations of ammonia and phosphorus in Wisconsin. Complementary evidence suggests that impacts may differ across production systems: prior work points to potentially larger externalities for hog

⁷In the days before the most extreme of such high precipitation events to hit North Carolina, namely Hurricane Florence in 2018, the handling of manure by hog operators was observed. Many illegally sprayed waste onto fields that were to be soaked in rain. This revealed at the same time how AFO operators may adopt averting behavior to prevent lagoons from overflowing, by preemptively emptying them — and thereby violating their permit — but yet do not prevent the manure from entering the water resources, due to saturated soils (Ouzts, 2018).

operations (Wing & Wolf, 2000) and indicates that measurable effects can also arise from facilities below the 1,000–animal unit cutoff. Methodologically, their use of Hydrologic Unit Codes (HUCs) provides a transparent and replicable spatial framework; at the same time, because HUC boundaries may not coincide with station-specific drainage basins (Omernik et al., 2017), estimates based on HUC aggregation may differ from those obtained using outlet-defined upstream catchments.

An equally abundant literature has documented links between AFOs and human health outcomes. Already in 2008, the Government Accountability Office flagged “15 studies that linked animal waste from industrial livestock farms with widespread health problems”(U.S. GAO, 2008). Since then, suggestive evidence has grown substantially, with numerous studies showing associations between AFO proximity and adverse conditions, including respiratory illness (Mirabelli et al., 2006; Poulsen, Pollak, Sills, Casey, Nachman, et al., 2018; Radon et al., 2007; Rasmussen et al., 2017; Sigurdarson & Kline, 2006), digestive disorders (Poulsen, Pollak, Sills, Casey, Rasmussen, et al., 2018), impaired mental health (Donham et al., 2007), and adverse birth outcomes such as low birth weight and infant mortality (Kravchenko et al., 2018).

While this literature suggests potentially important effects of AFOs on human health—from morbidity, including both mental and physical health, to mortality rates—through the environmental exposure pathway, the observed relationships are generally not causally interpretable due to selection effects. Households living near AFOs may differ systematically in other dimensions, such as socioeconomic status, baseline health, or other environmental risks, that make them at higher risk of deleterious health outcomes independently of AFO exposure, thereby confounding observed correlations. Two studies provide some causal evidence. Wing and Wolf (2000) use a matching design and find increased rates of several respiratory and gastrointestinal problems with proximity to swine operations, suggesting multiple contamination pathways including water exposure, though their analysis relies on a small sample (155 respondents). Sneeringer (2009) uses changes in county-level animal units to estimate impacts on infant health, and finds that increases in industrial production worsen infant morbidity and mortality, with evidence suggesting an air pollution channel, but is inconclusive regarding a potential water channel.⁸

Taken together, this evidence base suggests that AFOs may generate substantial environmental and health risks, but the magnitude and causal attribution remain uncertain. The current regulatory context, projected production growth, and the methodological limitations of prior work underscore the need for robust quantification of water pollution and its downstream health consequences.

3 Data

Animal Feeding Operations

I focus on two U.S. states with a high density of animal feeding operations, Iowa

⁸The analysis does not test for a statistical difference between the effects for counties with low/medium/high well usage, and does not have sufficient statistical power to detect an effect on the rates of infant death with causes related to water pollutants associated with livestock production.

and North Carolina—which together notably account for 31.4% and 12.3% of the U.S. swine inventory, respectively (USDA NASS, 2024). For both states, I compile and harmonize administrative records from the relevant state agencies.

These records report each operation’s location, start date, animal capacity by species, and waste management practices, including manure storage structures and acreage for land application. All operations are observed as geocoded point locations. I use reported animal capacities to compute livestock intensity in standardized animal units, which proxy for the quantity of manure generated—a quantity that is not directly observed—as is standard in the literature (Copeland, 2010). The permit databases do not include the locations of sprayfields. However, manure transport is costly, particularly for liquid manure, making it reasonable to assume that most land application occurs close to the waste storage structures. Survey evidence from Ali et al. (2012), based on livestock producers in Iowa and Missouri, indicates that among producers who transport manure off-site, the average maximum distance is below five miles.

The regulatory and institutional environments differ across the two states. In Iowa, construction permits are required for confinement feeding operations exceeding 1,000 animal units (about 2,500 adult pigs), and facility openings and expansions occur throughout the study period (2004–2021). In North Carolina, by contrast, a moratorium on new or expanded liquid-manure operations was imposed in 1997 and later made permanent for facilities using anaerobic lagoons. As a result, the number and spatial distribution of liquid-manure AFOs has remained largely unchanged since 1997. Permit requirements are also stricter than in Iowa, applying to all operations with more than 250 swine or 100 cattle that use some liquid waste management system, but exclude a large range of dry-poultry operations—which have substantially expanded in the post-moratorium period. To account for these operations, I therefore supplement the North Carolina administrative panel with geospatial predictions of unpermitted dry poultry operations derived from high-resolution remote sensing imagery (Handan-Nader & Ho, 2019), and restrict the North Carolina sample to the period post-2014, when coverage of dry poultry operations becomes available. Figure 1 shows the spatial distribution of the permitted AFOs in the top subfigure and the supplemented dataset in the bottom subfigure.

Figure 2 shows the resulting coverage of facilities in the two panels. To benchmark this coverage, I use the *NPDES CAFO Permitting Status Report*, published annually by the EPA since 2011. It compiles state-reported data and distinguishes between CAFOs (i.e., facilities above 1,000 animal units that fall under federal regulation) and those CAFOs that have actually obtained an NPDES permit. As required under 40 CFR 122.23(d)(1), only CAFOs that discharge must hold an NPDES permit, though states may impose additional requirements. Two patterns stand out. First, in both states there is a striking gap between the number of CAFOs with NPDES permits (light blue) and the total number of CAFOs (dark blue), suggesting that the overwhelming majority of federally regulated operations above 1,000 animal units operate without an NPDES permit. Second, my constructed panels provide extensive coverage across three policy-relevant categories: (i) federally regulated and permitted CAFOs, (ii) federally regulated but unpermitted CAFOs, and (iii) smaller AFOs below 1,000 animal units that fall outside federal regulation altogether.

Water quality outcomes

I collect water quality data from the Water Quality Portal ([Read et al., 2017](#)), which aggregates records from hundreds of institutions and agencies dating back to the early 20th century. The analysis aims to quantify changes in surface water quality attributable to upstream AFO activity, focusing on pollutants known to pose risks to public health and aquatic ecosystems. I therefore consider three categories of indicators: (1) pathogens originating from food animals and that can cause severe gastrointestinal illness, (2) nutrient pollution driven by nitrogen and phosphorus, and (3) dissolved oxygen as a general indicator of ecosystem health.

Because indicator availability, measurement frequency, and reporting conventions vary widely across pollutants and monitoring programs, I proceed in two steps. First, I identify a broad set of candidate indicators motivated by plausible environmental exposure pathways from AFOs. Second, I apply extensive data cleaning and harmonization procedures and screen outcomes using formal statistical power calculations, and retain only those indicators for which the panel structure and identifying variation support informative estimation. The raw data sample sizes and cleaning steps are documented in Appendix [A.1](#), and the power-screening procedure in Appendix [A.2](#).

Fecal coliform bacteria, which reside in the digestive tracts of food animals and are present in their manure, are pathogens that can induce severe gastrointestinal illness under short-term exposure ([U.S. EPA, 2009](#)). Their presence in surface waters is a strong indicator of fecal contamination from animal agriculture. The U.S. EPA has established a national primary drinking water regulation with a strict Maximum Contaminant Level (MCL) of zero detectable colonies for fecal coliforms (see Table [A2](#)).

Nutrient pollution—specifically nitrogen and phosphorus—is a major driver of surface water degradation, with consequences for ecosystem health, recreational use, and drinking water quality. In this analysis, I focus on aggregate indicators that capture overall nutrient loading. For nitrogen, I use Total Nitrogen (TN), which measures the sum of all nitrogen forms present in the water column, and Total Kjeldahl Nitrogen (TKN), which captures the combined organic and ammonia nitrogen fraction and is commonly used as an indicator of reactive nitrogen inputs from agricultural sources. For phosphorus, I consider Total Phosphorus (TP), which captures the overall phosphorus burden, including dissolved, particulate, and organic forms. While no drinking water standards regulate nutrient concentrations, the EPA has established recommended surface water quality criteria at the national level for phosphorus (0.1 mg/L in flowing waters, 0.05 mg/L in streams entering lakes) and ecoregion level for both N and P (see Table [A2](#)).

Finally, I extract measurements of dissolved oxygen (DO), which reflects the amount of oxygen available to aquatic life and serves as a general indicator of aquatic ecosystem health ([Keiser & Shapiro, 2019](#)). Using water temperature measurements taken within 24 hours of the DO sample, I convert DO readings into dissolved oxygen deficit, a normalized measure of oxygen depletion relative to saturation that provides a sharper indicator of eutrophication risk.

Figure 3 shows the locations of water monitoring stations with measurements of these water quality indicators, along with their number of observations, in both states over the respective study periods. Examples are shown for one indicator from each category.

Drainage areas

I construct the relevant spatial units for the analysis by delineating the drainage basin of each monitoring station. Using the National Hydrography Dataset Plus High Resolution (NHDPlus HR), which encodes flow direction for each stream segment of the U.S. surface hydrological network, I delineate the exact drainage basin for each monitoring station. This allows me to identify all AFOs located upstream of a given station and to measure their distances to the station outlet. By construction, runoff from these facilities flows downslope into the monitored stream segment. Importantly for identification, drainage basin boundaries are orthogonal to administrative units such as counties, along which AFOs may choose to locate.

Extreme precipitation events

I use daily precipitation data from the Parameter-elevation Regressions on Independent Slopes Model (PRISM) dataset, which provides 4-km resolution coverage for the entire United States from 1981 onward (PRISM, 2020). These data are used both as controls for local hydrologic conditions and to identify extreme precipitation events for the event-study analysis. To define extremes, I process the data in two steps designed to preserve the relevant dimension of weather variation. First, I compute the distribution of daily precipitation at each grid cell over a 30-year reference period (1981–2010), following World Meteorological Organization guidelines (WMO, 2017). I define an extreme precipitation event as daily rainfall exceeding two standard deviations above the seasonal mean at that location. Seasons are defined using three-month groupings (JFM, AMJ, JAS, OND). An extreme precipitation event is thereby defined relative to the local climatology: specifically, rainfall that is substantially higher than what is normal for that location, season, and period. Second, I aggregate extreme events across all upstream AFO locations within a drainage area.

Controls

To account for confounding from other land-based pollution sources and hydrologic conditions, I include a rich set of time-varying controls.

Land cover data are drawn from the National Land Cover Database (NLCD), which provides 30-meter resolution classifications for the contiguous U.S. (Dewitz & USGS, 2021). Using annual NLCD layers from 1985 to 2023, I compute for each drainage area and year the total shares of land classified as planted, developed, wetlands, and other. These variables capture notably the presence of crop agriculture, of hydrologically sensitive areas, and the potential for other non-point source pollution.

To further control for non-manure nutrient inputs, I use county-level estimates of the share of cropland receiving commercial fertilizer from the U.S. Census of Agriculture. I

linearly interpolate values between census years and reallocate county-level fertilizer shares to drainage areas in proportion to the share of each drainage area’s surface lying within each county.

I also control for total precipitation falling within each drainage area (and its square). Finally, because water temperature affects biological activity and nutrient dynamics, I include it as a control variable, except in regressions where dissolved oxygen deficit is the outcome, as this measure already accounts for thermal conditions.

Final samples

The final estimation samples are defined at the station–day level and include all dates on which valid measurements are available for water quality indicators retained after data cleaning and power screening. Summary statistics for the main outcomes, exposure variables, and controls are reported separately for Iowa and North Carolina in Table 1.

Several differences between the two state settings are noteworthy. Drainage areas of monitoring stations are substantially larger on average in North Carolina than in Iowa, reflecting differences in hydrology and topography. Correspondingly, upstream exposure to animal production differs in scale: while Iowa drainage areas are characterized by lower raw counts of AFOs, North Carolina drainage areas typically encompass more upstream facilities. However, the average number of AUs per drainage area is lower in North Carolina, indicating a higher frequency of smaller facilities.

Water quality levels also differ markedly across states. Average concentrations of fecal coliforms are substantially higher in Iowa surface waters than in North Carolina, indicating large differences in pathogen loads across settings. Nutrient concentrations show similar contrasts: both nitrogen and phosphorus indicators exhibit higher mean levels in Iowa, although considerable within-state variation is present in both panels. In both states, average total phosphorus concentrations—in both whole-water and dissolved samples—exceed national and ecoregion-specific recommended surface water quality criteria (see Table A2).

4 Empirical strategy

Notation and exposure construction

For each monitoring station i with measurements of water quality on day t , let $\{U_j\}_{j=1}^{J[it]}$ denote the set of AFOs located upstream of i within its delineated drainage area. Let d_{ij} denote the distance between station i and upstream facility j . Let AU_{jt} denote the permitted animal-unit capacity of facility j at time t . Finally, let E_{jt} denote an indicator for the occurrence of an extreme precipitation event at facility j on day t .

Using these primitives, I construct measures of upstream exposure by aggregating across all upstream facilities, either as raw counts or as inverse-distance-weighted sums, with weights proportional to $(1+d_{ij})^{-1}$. When relevant, exposure is further decomposed by facility size (above versus below 1,000 animal units) or by production measure (AFO counts versus animal units).

Identification strategies

The ideal research design, from a statistical standpoint, would be a randomized experiment in which the opening or the expansion of an AFO is randomly assigned in time and space, while all other characteristics are held constant. This would ensure that variation in pollution exposure is independent of factors impacting the outcomes of interest. In practice, I rely on two complementary sources of quasi-experimental variation: changes in upstream livestock intensity over time, which capture chronic pollution effects, and exogenous precipitation shocks at facility locations, which capture acute pollution responses.

In Iowa, where the number and sizes of AFOs vary over time and across space during the study period, I adopt a difference-in-differences design with continuous treatment intensity. Specifically, I exploit spatio-temporal variation in upstream animal production within drainage areas to estimate the causal effect of changes in livestock intensity on downstream surface water quality. Identification relies on a conditional parallel trends assumption, namely that absent changes in animal production, water quality outcomes would have followed similar trends across drainage areas.

In both Iowa and North Carolina, I also implement an event-study design based on extreme precipitation events at the locations of upstream facilities. Heavy rainfall increases the likelihood that stored or land-applied manure enters surface waters through overflow, runoff, or leaching, while being plausibly orthogonal to changes in livestock production itself. This design identifies acute pollution effects driven by exogenous shocks to pollutant transport rather than by changes in production intensity. In my main specifications, the treatment variable is the sum of daily precipitation shocks experienced by upstream AFOs over the seven-day window preceding measurement of the outcome.

Difference-in-differences framework (chronic effects)

To estimate the chronic effects of upstream animal production on downstream surface water quality, I estimate panel regressions of the form (D0)

$$\text{WQ}_{it}^k = \mathbf{A}\mathbf{P}'_{it}\boldsymbol{\beta} + X'_{it}\gamma + \alpha_i + \omega_t + \varepsilon_{it}, \quad \forall i = 1, \dots, N; t = 1, \dots, T \quad (\text{D0})$$

where WQ_{it}^k denotes the average level of water quality indicator k measured at station i on day t . The vector $\mathbf{A}\mathbf{P}_{it}$ is a vector-valued function of upstream animal production exposure, constructed by aggregating across all animal feeding operations located in the drainage area of station i . Depending on the specification, animal production can be measured either as counts of upstream operations (AFOs) or as total upstream animal units (AUs). The parameters of interest are the elements of $\boldsymbol{\beta}$, which capture how downstream water quality responds to changes in upstream animal production under the specified exposure mapping. Station fixed effects α_i absorb time-invariant differences in hydrology and baseline land use, while time fixed effects ω_t (year and month) flexibly control for seasonality and common shocks and state-wide factors evolving over time, such as policies targeting non-point source water pollution, respectively.

Rather than imposing a single mapping from upstream production to pollution exposure, I estimate a set of specifications that each isolate a distinct dimension of the relationship

between livestock production and water quality. Together, these specifications characterize how pollution responds to proximity, operation size, production structure, and allow for non-linear exposure–response relationships. Specifically, I estimate the following difference-in-differences specifications, which differ in how upstream production intensity enters the regression through $\mathbf{AP}'_{it}\boldsymbol{\beta}$:

(D1) **Raw count of upstream operations.** $\mathbf{AP}'_{it}\boldsymbol{\beta} := \beta \text{AFO}_{it}$

This baseline specification uses the total number of upstream animal feeding operations and estimates the average change in downstream water quality associated with adding one additional operation within the drainage area.

(D2) **Inverse-distance-weighted exposure (IDW).** $\mathbf{AP}'_{it}\boldsymbol{\beta} := \beta_d \text{AP}_{it}^d$

This specification replaces the raw count with an inverse distance-weighted measure of upstream exposure, allowing the impact of animal production to decay continuously with hydrological distance from the monitoring station. The coefficient captures the effect of increasing *effective* upstream exposure by one unit of distance-weighted production.

(D3) **Nonlinear exposure–response relationship.** $\mathbf{AP}'_{it}\boldsymbol{\beta} := \beta_l \text{AP}_{it}^d + \beta_q \text{AP}_{it}^{d^2}$

To allow for nonlinearities, I include both linear and quadratic terms in the IDW exposure. This specification tests whether marginal pollution effects vary with the level of upstream animal production, distinguishing saturation effects from amplification at higher exposure levels.

(D4) **Operation size: facilities above versus below the federal regulatory threshold.** $\mathbf{AP}'_{it}\boldsymbol{\beta} := \beta_a \text{AP}_{it}^{\geq 1,000,d} + \beta_b \text{AP}_{it}^{< 1,000,d}$

This specification decomposes upstream exposure by operation size to assess whether facilities below and above the 1,000–animal-unit threshold have systematically different marginal impacts on downstream water quality.

(D5) **Production scale versus concentration.** $\mathbf{AP}'_{it}\boldsymbol{\beta} := \beta_s \text{AFO}_{it}^d + \beta_c \text{AU}_{it}^d$

Finally, by jointly including the number of upstream operations and total upstream animal units, this specification isolates whether pollution responds to total livestock burden or to how that burden is distributed across more or fewer operations.

Extreme precipitation framework (acute effects)

To estimate the acute effects of upstream animal feeding operations on downstream surface water quality, I relate water quality outcomes to extreme precipitation events occurring at the locations of upstream facilities. I estimate regressions of the form (P0)

$$\text{WQ}_{it}^k = \mathbf{PE}'_{it}\boldsymbol{\theta} + X'_{it}\boldsymbol{\gamma} + \alpha_i + \omega_t + \varepsilon_{it}, \quad \forall i = 1, \dots, N; t = 1, \dots, T \quad (\text{P0})$$

where $\mathbf{PE}_{it}$ is a vector aggregating extreme precipitation events occurring at animal feeding operations upstream of station i during the week(s) preceding day t . The coefficients $\boldsymbol{\theta}$ capture the reduced-form effect of extreme precipitation at upstream animal feeding operations on downstream water quality. Because precipitation shocks are transitory, these estimates are

interpreted as short-run impacts associated with acute release or mobilization of pollutants rather than with changes in production intensity. The vector of controls X_{it} additionally includes the number of upstream AFOs to absorb baseline exposure levels. Other controls and fixed effects follow the same structure as in the difference-in-differences framework.

Here as well, rather than imposing a single aggregation of precipitation shocks, I estimate a set of specifications that isolate distinct dimensions of acute exposure, including proximity, nonlinearity, and temporal persistence. Specifically, I estimate the following precipitation-event specifications, which differ in how extreme precipitation at upstream facilities enters the regression through $\mathbf{PE}'_{it}\boldsymbol{\theta}$:

(P1) **Raw count of upstream precipitation events.** $\mathbf{PE}'_{it}\boldsymbol{\theta} := \theta \sum_j E_{jt}$

This baseline specification uses the total number of extreme precipitation events affecting upstream facilities in the seven days preceding t and estimates the average short-run change in downstream water quality associated with one additional event.

(P2) **Inverse-distance-weighted exposure.** $\mathbf{PE}'_{it}\boldsymbol{\theta} := \theta_d \sum_j \frac{1}{1+d_{ij}} E_{jt} \equiv \theta_d \mathbf{PE}_{it}^d$

This specification weights precipitation events by inverse hydrological distance, allowing events occurring closer to the monitoring station to have a larger impact on measured water quality.

(P3) **Nonlinear distance-weighted exposure.** $\mathbf{PE}'_{it}\boldsymbol{\theta} := \theta_l \mathbf{PE}_{it}^d + \theta_q \mathbf{PE}_{it}^{d^2}$

By including both linear and quadratic terms in distance-weighted exposure, this specification allows for non-linear functional forms of the marginal pollution response.

(P4) **Temporal persistence of precipitation effects.** $\mathbf{PE}'_{it}\boldsymbol{\theta} := \theta_{L_0} \mathbf{PE}_{it}^d + \theta_{L_1} \mathbf{PE}_{it}^{L_1, d}$

This specification separates week-of and further lagged precipitation exposure, allowing acute pollution responses to differ between recent runoff effects and delayed transport or persistence in surface waters—from events occurring up to 14 days prior to the measurement of the outcome.

Weights and inference

The daily frequency of water quality monitoring varies across stations and sampling dates. To account for differences in measurement precision, I weight all regressions by the number of underlying daily observations at station i used to construct the daily average outcome on day t . This weighting scheme gives greater influence to observations based on more raw field measurements and mitigates heteroskedasticity arising from uneven sampling effort.

Inference is conducted using standard errors clustered at the drainage-area level, which is the level of identifying variation in both the difference-in-differences and precipitation-event designs. This allows for arbitrary correlation of errors within watersheds over time.

Because the combination of fixed effects, clustering, and irregular sampling substantially reduces the effective sample size, not all water quality indicators can be analyzed with adequate statistical power. I therefore conduct power simulations under the baseline, least-demanding specification in each design framework and restrict the analysis to indicators for which the design is not clearly underpowered. This screening avoids interpreting estimates

from specifications that are highly unlikely to be informative, while not ruling out the possibility that some remaining specifications may still be underpowered.

Restricted sample to account for upstream-downstream connectivity

The data generating process of water quality levels follows a specific upstream - downstream structure. Some monitoring stations are nested within the drainage areas of others, such that their pollutant readings are partly determined by the levels experienced at upstream stations whose drainage areas are subsumed under the larger area of the downstream station. When two such hydrologically connected stations report measurements of the same parameter in the same time period, including both observations without adjustment can mechanically induce dependence across regression residuals and bias inference.

To address this issue, I identify the full hydrological network of monitoring stations—the “watershed matrix”⁹—which delineates upstream-downstream connectivity across stations. I then restrict the estimation sample so that, for each water quality indicator k and day t , only stations that are not downstream of another station reporting measurements of k at t are retained. In other words, I exclude ‘nesting’ observations. This restriction limits dependence induced by upstream-downstream nesting and ensures that station-time observations used in the estimation are hydrologically independent.

5 Results

Results are presented in Tables 2 through 5. From top to bottom, each table reports (i) the estimated *level effects*, i.e. raw regression coefficients with clustered standard errors, (ii) the *sample mean* of the outcome within the regression sample, and (iii) the corresponding *relative effects*. The latter are additive level effects expressed as percent-of-mean changes. For specifications that are linear in exposure, the relative effect is computed as the estimated coefficient divided by the sample mean of the outcome. For specifications that include quadratic terms, the total reported relative effect is based on the marginal effect evaluated at the sample mean of the (non-squared) exposure, which is then normalized by the outcome mean. This presentation allows direct comparison of magnitudes across specifications and outcomes.

At the bottom of each table, additional summary statistics are reported to aid interpretation. In particular, the row labeled *Cluster variation* indicates whether the identifying variation in the corresponding regression is supported by a sufficient number of drainage-area clusters that exhibit within-cluster variation in treatment after accounting for fixed effects. This diagnostic is relevant because observations belonging to clusters with no such variation do not contribute identifying information under cluster-robust inference, even though they

⁹Given a spatial directed network of J units, the $J \times J$ watershed matrix characterizes all adjacent and non-adjacent connectivity between units. In our setting, a value of 1 at row i and column j indicates that station i is located upstream from station j , although multiple other stations may be located at intermediary locations along this flow path, such that the water entering station i drains into other stations before entering station j . The watershed matrix thereby accounts for the full downstream connectivity of subwatersheds. It can be computed from the more common adjacency matrix, which only indicates adjacent connections.

enter the nominal sample. Reporting this indicator helps distinguish specifications where estimated effects are supported by a meaningful set of informative clusters from those where effective identifying variation is limited.

5.1 Chronic contamination from AFO activity

Tables 2 and 3 report difference-in-differences estimates for Iowa, exploiting spatial and temporal variation in upstream livestock intensity to identify chronic effects of animal production on downstream surface water quality.

The first baseline specification in Table 2 relates outcomes to the raw count of upstream AFOs. This specification implicitly assigns equal weight to all upstream facilities, regardless of their distance from the monitoring station. Because such a mapping does not reflect how pollutants are transported in surface waters, the baseline results are primarily reported for reference and are not emphasized. Accordingly, all subsequent specifications rely on inverse-distance-weighted (IDW) exposure measures that allow pollution impacts to decay with distance.

For all relevant specifications, two versions are estimated: one measuring upstream production as the number of facilities, and the other—indicated by a prime—measuring production as total upstream animal units. In the latter case, coefficients are scaled so that reported effects correspond to an increase of 1,000 animal units rather than a single unit, ensuring comparability of magnitudes across formulations.

Under IDW exposure (D2), increases in upstream livestock intensity are associated with statistically significant increases in dissolved oxygen deficit and total Kjeldahl nitrogen. These effects are similar in magnitude when exposure is measured using facility counts or animal units, while dissolved total phosphorus responds when exposure is expressed in animal units. Allowing for nonlinearities (D3) yields evidence of a concave exposure–response relationship in the case of oxygen deficit, with positive marginal effects diminishing as upstream intensity increases. This pattern is consistent with the bounded nature of dissolved oxygen deficit and with nonlinear biogeochemical and physical processes governing oxygen dynamics in streams. As organic matter and nutrient loading increase, oxygen consumption initially rises sharply, but marginal changes in measured deficit diminish at higher exposure levels.

Table 3 reports specifications that focus on two additional dimensions of upstream exposure. Specification (D4) decomposes distance-weighted exposure by facility size relative to the 1,000–animal-unit regulatory threshold. When exposure is measured using facility counts, the estimated effects are driven primarily by large operations, which is consistent with the fact that these facilities account for a disproportionate share of total production. However, in the animal-unit formulation, the most frequently measured outcome—oxygen deficit—also responds positively to animal units at facilities below the threshold, indicating that smaller operations contribute to chronic pollution when considered in terms of livestock mass.

The distribution of permitted animal units exhibits visible bunching just below the 1,000–AU threshold in the administrative data. To account for this, I conduct a robustness check analysis that re-estimates specification (D4) using the alternative threshold of 996 animal units. The estimated effects are qualitatively similar under this alternative definition, supporting the robustness of the main results (see Appendix Table A3).

Specification (D5) finally isolates the role of production scale versus concentration by jointly including distance-weighted facility counts and distance-weighted animal units. In this specification, the coefficient on animal units captures the effect of increasing total upstream livestock mass holding the number of facilities fixed, while the coefficient on facility counts captures the effect of changing the number of operations holding total animal mass constant, i.e., the effect of changes in the concentration of animals. Focusing on dissolved oxygen deficit—the most precisely estimated outcome and therefore the setting in which the two correlated exposure measures can be most informatively disentangled—the joint estimates indicate that, holding total upstream animal units fixed, variation in the number of upstream facilities has little effect on oxygen deficit. In contrast, holding the number of facilities constant, increases in total animal units are associated with higher oxygen deficit. This pattern implies that chronic oxygen depletion scales with total livestock load rather than with how that load is distributed across operations. Total phosphorus similarly responds to total animal units, while the coefficient on facility counts enters with the opposite sign, suggesting that, conditional on total livestock mass, having smaller operations may be associated with lower phosphorus concentrations than configurations with fewer, bigger AFOs.

Taken together, the difference-in-differences results indicate that chronic downstream water quality degradation in Iowa is closely related to upstream livestock intensity when exposure is mapped through hydrologically relevant, proximity-weighted measures. The most robust and precisely estimated effects are observed for dissolved oxygen deficit, which responds systematically to increases in total upstream animal units and exhibits a concave exposure–response relationship. Nitrogen outcomes display qualitatively similar patterns, though with less precision in nonlinear specifications. Across specifications that disentangle production scale from production structure, the evidence points to total livestock load as the primary driver of chronic impacts, with the most precisely estimated results providing no indication that concentrating animals in fewer, denser operations reduces pollution and, if anything, suggesting lower nutrient pollution levels when a given livestock burden is distributed across more numerous, smaller facilities.

5.2 Acute contamination from extreme precipitation events

Tables 4 and 5 report event-study estimates linking downstream water quality to extreme precipitation events occurring at upstream animal feeding operations, in Iowa and North Carolina, respectively. These specifications identify short-run, acute responses driven by exogenous shocks to pollutant transport rather than by changes in livestock production.

In Iowa (Table 4), specifications based on raw counts of upstream precipitation events yield small but statistically significant relative changes in oxygen deficit and several nutrient measures. Once events are weighted by inverse distance, the estimated effects become substantially larger and statistically significant across a broad set of outcomes. Distance-weighted precipitation exposure is associated with increases in dissolved oxygen deficit as well as in nitrogen and phosphorus concentrations, including both total and dissolved forms. Specifications that separate contemporaneous and lagged precipitation exposure show that acute effects are not confined to the immediate week of the event: for some indicators, particularly oxygen deficit and total nitrogen, lagged exposure remains relevant, while for others

the response is concentrated in the seven days preceding measurement of the outcome.

The North Carolina results in Table 5 display a closely aligned pattern despite differences in geography, production systems, and monitoring regimes. Distance-weighted precipitation events are associated with statistically significant increases in dissolved oxygen deficit and total phosphorus, and with positive responses in nitrogen measures in several specifications. A distinctive feature of the North Carolina analysis is the measurement of fecal coliforms, which show a clear pollution pattern. Fecal coliforms rise sharply following distance-weighted precipitation exposure and exhibit a concave functional form. This outcome provides direct evidence of short-run microbial contamination following extreme rainfall at upstream facilities. Allowing for nonlinearities indicates concave exposure–response relationships for selected outcomes, with positive marginal effects that diminish at higher exposure levels, and lagged precipitation still has impacts on DO deficit.

As a robustness check, I consider an alternative definition of precipitation shocks that considers the week as the relevant reference period to identify anomalous precipitation levels. In these specifications, extreme precipitation is defined based on the total amount of rainfall received over the previous week relative to the climatological distribution for that location, i.e., extreme week events are defined as weeks in which total precipitation exceeds two standard deviations of the historical week-level mean. The corresponding treatment variable then captures the number of upstream AFOs exposed to such extreme cumulative rainfall over the period. Appendix Tables A4 and A5 report the results obtained using these alternative measures, which are qualitatively similar to the baseline event-study estimates and support the robustness of the acute contamination findings.

Together, the event-study results from both states provide consistent evidence that extreme precipitation at upstream AFO locations leads to acute deterioration in downstream surface water quality. The effects are detected across indicators and are strongest when exposure is measured using hydrologically relevant proximity weights. These findings highlight the role of extreme rainfall as a trigger for short-run pollutant mobilization from animal production sites into surface waters.

6 Discussion

This study combines two complementary quasi-experimental identification strategies, tailored to the regulatory and data contexts of Iowa and North Carolina, to estimate the reduced-form effects of industrial animal production on downstream surface water quality. In Iowa, I exploit spatio-temporal variation in upstream livestock intensity—measured using both facility counts and animal units—between 2004 and 2021. In both Iowa and North Carolina, I additionally leverage plausibly exogenous extreme precipitation shocks at upstream animal feeding operations to identify acute pollution responses driven by short-run mobilization rather than by changes in production. Both approaches are motivated by mechanisms highlighted in the literature: chronic contamination driven by sustained manure loading, and acute contamination triggered when heavy rainfall increases the likelihood that stored or land-applied waste reaches surface waters.

Across both states, I find clear acute effects of extreme precipitation events at upstream facilities on downstream water quality, consistent with short-run mobilization of manure-related pollutants. In both Iowa and North Carolina, distance-weighted precipitation exposure increases dissolved oxygen deficit and nutrient concentrations, and in North Carolina is additionally associated with sharp increases in fecal coliforms, providing direct evidence of microbial contamination following heavy rainfall. The similarity of these responses across states indicates that extreme precipitation acts as a common trigger for downstream contamination, despite differences in production systems, species composition, and regulatory environments. Together, these results highlight both the chronic and acute dimensions of water pollution from intensive animal production.

This analysis makes important contributions along several dimensions. First, by combining data from Iowa and North Carolina, it provides multi-design, multi-state evidence across multiple livestock species and measures of production intensity, ranging from facility counts to animal units. Second, it shows that animal feeding operations affect water quality through both chronic pathways, linked to sustained livestock intensity, and acute pathways, activated by extreme precipitation shocks. Third, because the data include a large number of facilities below the federal CAFO threshold of 1,000 animal units—alongside nearly all facilities above it—the analysis detects pollution effects from operations that fall outside federal permitting requirements yet account for the majority of livestock sites. Finally, by explicitly comparing facility counts and animal units, the results suggest that pollution impacts scale with total animal mass, and that greater concentration of animals within fewer operations may actually exacerbate nutrient pollution relative to a distribution across smaller facilities.

Quantifying the pollution caused by AFOs is particularly timely, as the number of industrial animal operations and animal density within such operations continue to increase in the U.S. and abroad. In North Carolina, while swine AFOs have remained capped since the 1997 moratorium, poultry operations have expanded rapidly, raising new concerns about nutrient and pathogen loading. At the same time, climate change is projected to increase the frequency of extreme precipitation events,¹⁰ further elevating the risk of acute contamination from existing facilities.

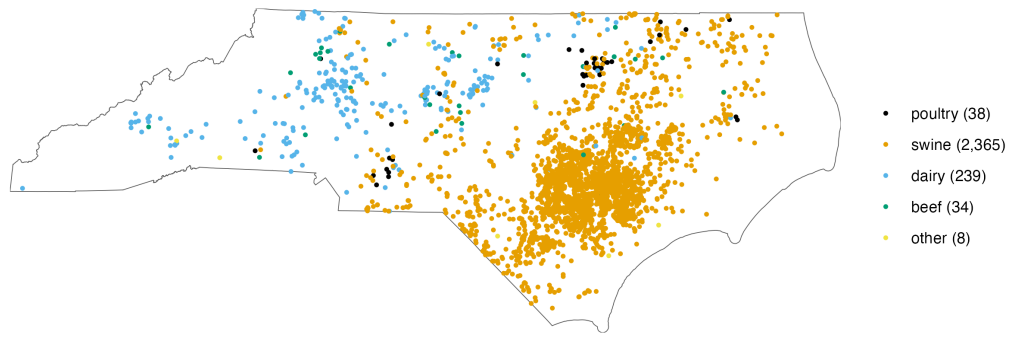
The estimates of surface water pollution obtained in this study underscore the relevance of ecosystem carrying capacity in the context of intensive livestock production. They suggest that the spatial concentration of farm animals—and therefore of manure—at industrial scales may exceed the assimilative capacity of local landscapes and waterways, such that sprayfields and surface waters are unable to retain or sufficiently dilute released contaminants. Similarly, Joy et al. (2022), in the context of dairy operations in New Zealand, estimate farms’ nitrate grey water footprint—the volume of water required to dilute nitrogen leaching to meet water quality standards—and find that maintaining healthy water

¹⁰The state’s latest climate science report, which defines a “heavy rainfall event” for North Carolina as a day on which rainfall totals 3 inches (76.2 mm) or more, reports an observed upward trend in the number of such events, with the period 2015-2018 having seen the greatest number of events since 1900. Its projections for the end of the century state that “it is *likely* that annual total precipitation for North Carolina will increase”, and “*very likely* that extreme precipitation frequency and intensity in North Carolina will increase”, due to increases in atmospheric water vapor content (Kunkel et al., 2020).

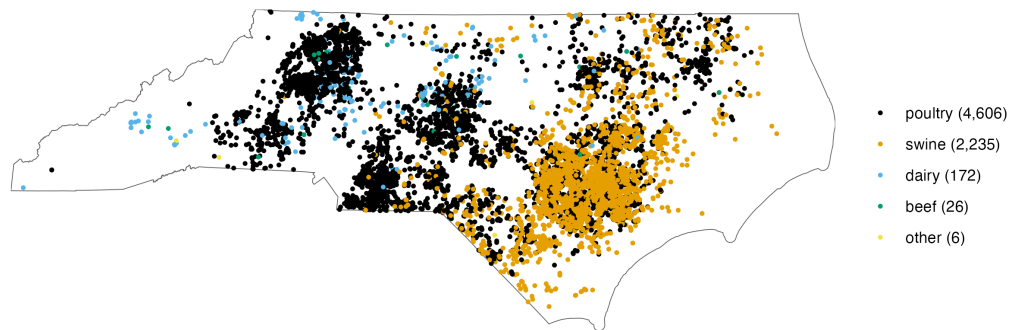
under existing production levels would require either a twelve-fold increase in rainfall or a twelve-fold reduction in herd size. In this sense, the present results point to comparable constraints in U.S. livestock regions: given the high costs of manure transport and limited scope for redistribution across space, current patterns of livestock density may be environmentally unsustainable in the absence of substantial changes in production practices or scale.

These results open up an array of questions that are beyond the scope of this study and call for further research. In particular, the present analysis cannot distinguish whether observed pollution primarily originates from structural failures or overflow at manure storage facilities or from runoff and leaching following land application on sprayfields. Data on the precise spatial extent and timing of sprayfield applications—while often reported in permit files—are not systematically digitized and therefore cannot be incorporated here. Access to such data would allow a more precise attribution of pollution sources and would facilitate the design of targeted, short-term policy interventions to limit downstream contamination.

The research designs employed here also capture only one dimension of water pollution from animal operations, namely contamination of surface waters. However, pollutants may also enter groundwater systems, affecting downstream populations through this channel, particularly those relying on domestic wells for drinking water. Indeed, localized studies have documented elevated contaminant levels in both surface water and groundwater downstream from sprayfields ([Harden, 2015](#); [Karr et al., 2001](#)), though such impacts have not yet been robustly assessed using causal research designs. An estimated 37.3 million individuals relied on domestic well water in the contiguous U.S. in 2010 ([Johnson et al., 2019](#)), and North Carolina has the largest number of housing units served by private wells, exceeding 1.5 million ([Murray et al., 2021](#)). The consistent evidence of increased downstream surface water contamination documented in this study suggests that groundwater impacts may also be present, with potentially serious implications for human health, highlighting the need for future work that jointly examines surface and subsurface pathways of exposure.



(a) Registered AFOs (1997-2020 database)



(b) Registered AFOs (2014-2020 database) and poultry facilities from remote sensing imagery (2014)

Figure 1: Location of AFOs in North Carolina by animal type

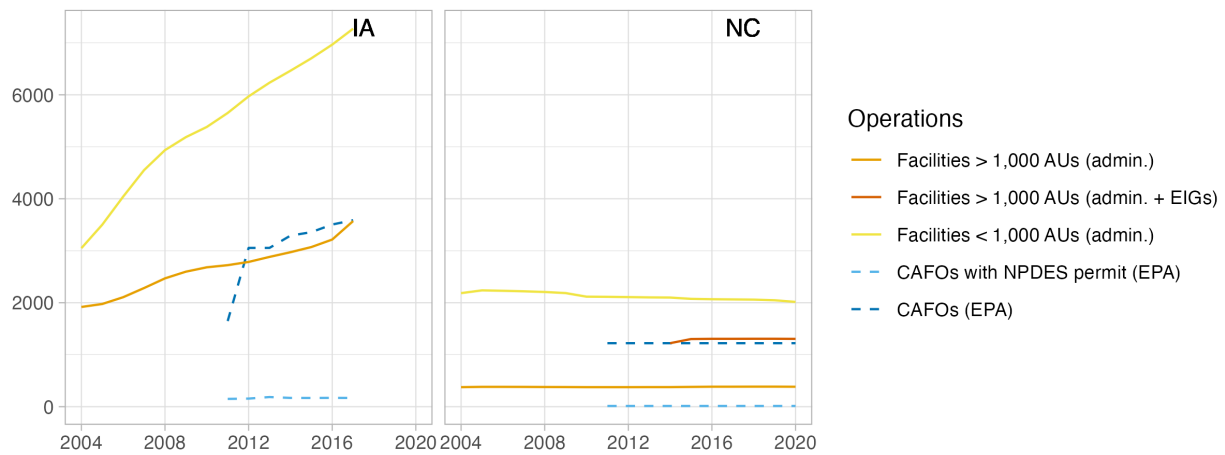


Figure 2: Coverage of the state AFO panels wr.t. size and regulation: total AFO counts by category. Notes: The red line corresponds to the addition of the manually validated poultry operations from remote sensing imagery

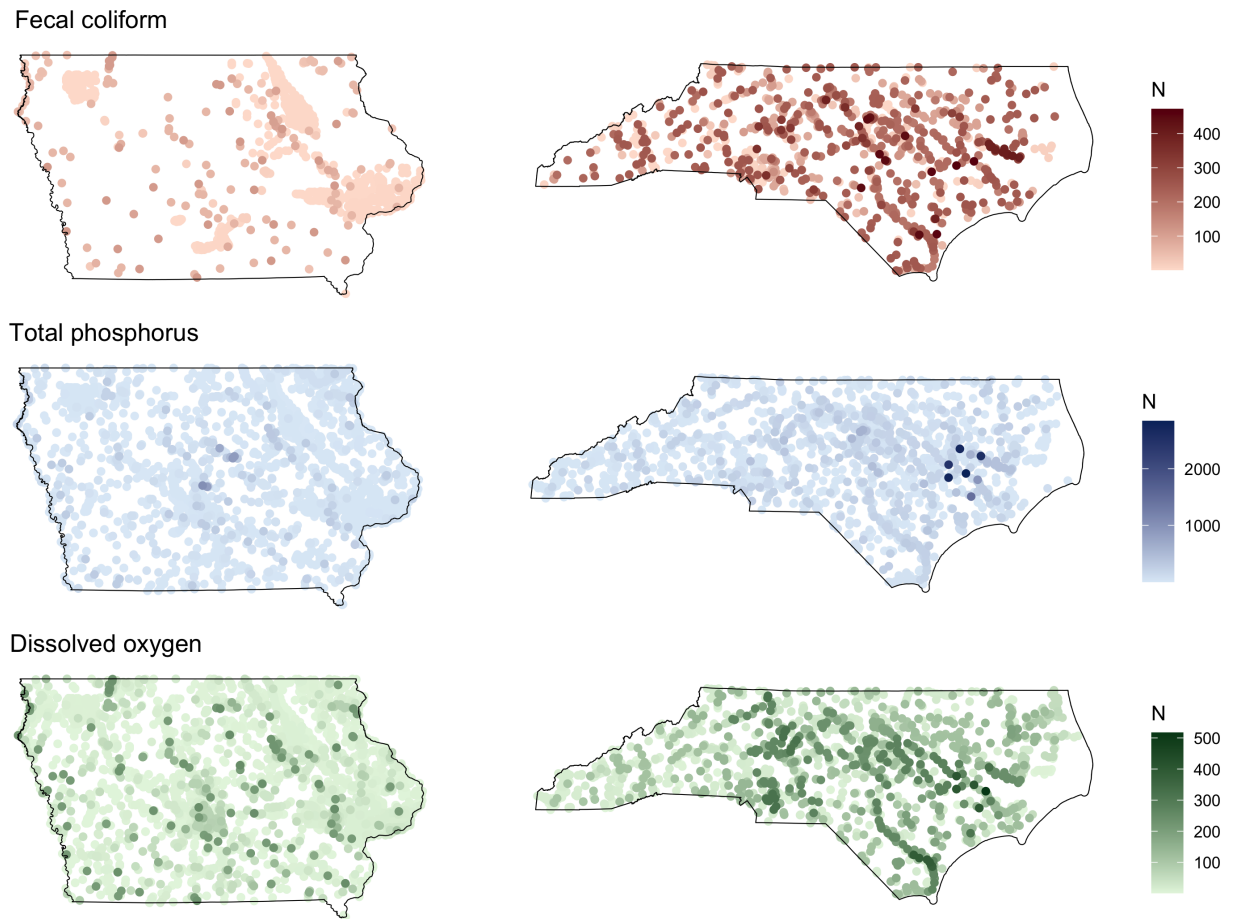


Figure 3: Location and number of observations of water monitoring stations, in Iowa (2004-2017) and North Carolina (1997-2020), with measurements of fecal coliforms (top), total phosphorus (middle) and dissolved oxygen (bottom), respectively.

Variable	Unit	IA (2004-2021)		NC (2014-2021)	
		Mean	SD	Mean	SD
<i>Water quality indicators</i>					
DO deficit	%	15.4	21.0	15.2	20.2
Fecal coliforms	cfu/100mL	2,785	11,030	368	820
TKN _{dissolved}	mg/L as N			0.312	0.205
TKN _{whole}	mg/L as N	0.875	1.07	0.587	0.421
TN _{dissolved}	mg/L as N	7.33	4.73	0.938	0.481
TN _{whole}	mg/L as N	8.47	6.04	0.966	0.676
TP _{dissolved}	mg/L as P	0.157	0.265	0.045	0.031
TP _{whole}	mg/L as P	0.292	0.536	0.108	0.138
<i>Explanatory variables</i>					
AFO		47.4	117.	54.	155.
AFO ^d		1.67	2.58	2.46	4.98
AU		57,060	143,100	36,970	105,900
AU ^d		1,963	3,378	1,691	3,465
T _{water}	°C	13.6	7.3	16.6	7.0
Precipitation	mm	20.6	24.6	26.8	32.6
% _{crop land} fertilized	%	73	7	68	11
% _{land cover} developed	%	3	2	22	24
% _{land cover} planted	%	59	8	21	17
% _{land cover} wetlands	%	2	2	8	13
Drainage area size	km2	284	959	856	2,375

Table 1: Sample summary statistics. All outcome and explanatory variables are at the drainage-area-by-week level. The table reflects the pooled sample used across all water-quality regressions. Variables with a *d* superscript are inverse-distance weighted. Nutrients are indicated alongside their sample fraction.

Iowa — DiD analysis — Specifications (D1)-(D3)

Spec.	X	DO deficit	TKN _{whole}	TP _{whole}	TP _{diss.}
Level effect					
(1)	AFO	0.024*** (9.2E-3)	4.4E-4** (2.2E-4)	-2.2E-4 (1.6E-4)	1.7E-4 (1.8E-4)
(2)	AFO ^d	0.951*** (0.345)	0.026** (0.012)	-2.9E-3 (5.7E-3)	0.012 (0.012)
(2')	AU ^d	0.863*** (0.262)	0.027*** (9.5E-3)	2.0E-3 (6.8E-3)	0.03*** (0.011)
(3)	AFO ^d	1.83*** (0.624)	0.017 (0.042)	-6.9E-3 (0.016)	0.044 (0.033)
	AFO ^{d2}	-0.054** (0.023)	4.4E-4 (1.5E-3)	2.2E-4 (6.1E-4)	-2.6E-3 (2.6E-3)
(3')	AU ^d	1.68*** (0.568)	0.021 (0.032)	-2.1E-3 (0.019)	0.042* (0.025)
	AU ^{d2}	-4.6E-5** (0.021)	2.8E-7 (1.2E-3)	2.1E-7 (7.1E-4)	-1.3E-6 (2.2E-3)
Sample mean					
		15.4	0.875	0.292	0.157
Relative effect					
(1)	AFO	+0.16 %	+0.05 %	-0.08 %	+0.11 %
(2)	AFO ^d	+6.2 %	+2.9 %	-0.99 %	+7.5 %
(2')	AU ^d	+5.6 %	+3.1 %	+0.69 %	+19 %
(3)	AFO ^d +AFO ^{d2}	+11 %	+2.1 %	-2.1 %	+22 %
(3')	AU ^d +AU ^{d2}	+9.7 %	+2.5 %	-0.44 %	+24 %
Observations		31,382	13,648	13,696	946
% obs. below QL		0 %	4 %	0 %	7 %
Cluster variation		✓	✓	✓	
Power		✓	✓	✓	✓

Table 2: Effects of animal production on downstream surface water quality in Iowa, 2004–2021 — Difference-in-differences specifications (D1)-(D3). Variables with a *d* superscript are inverse-distance weighted. Standard errors are clustered by drainage area. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimates significant at the 5% level are shown in bold.

Iowa — DiD analysis — Specifications (D4)-(D5)

Spec.	X	DO deficit	TKN _{whole}	TP _{whole}	TP _{diss.}
Level effect					
(4)	AFO $\geq 1,000, d$	4.38***	0.084**	-3.8E-3	0.122**
		(1.02)	(0.037)	(0.026)	(0.058)
	AFO $< 1,000, d$	0.024	9.4E-3	-2.7E-3	-0.011
		(0.382)	(0.016)	(7.7E-3)	(0.015)
(4')	AU $\geq 1,000, d$	0.643**	0.034**	7.7E-3	0.063***
		(0.295)	(0.017)	(0.011)	(0.014)
	AU $< 1,000, d$	1.37**	0.014	-7.4E-3	-0.018
		(0.623)	(0.034)	(0.013)	(0.024)
(5)	AFO ^d	0.30	-5.2E-3	-0.02	-0.093***
		(0.452)	(0.021)	(0.015)	(0.027)
	AU ^d	0.67**	0.031*	0.018	0.105***
		(0.34)	(0.017)	(0.015)	(0.023)
Sample mean					
		15.4	0.875	0.292	0.157
Relative effect					
(4)	AFO $\geq 1,000, d$	+28 %	+9.6 %	-1.3 %	+78 %
	AFO $< 1,000, d$	+0.2 %	+1.1 %	-0.9 %	-6.9 %
(4')	AU $\geq 1,000, d$	+4.2 %	+3.9 %	+2.6 %	+40 %
	AU $< 1,000, d$	+8.9 %	+1.6 %	-2.5 %	-11 %
(5)	AFO ^d	+2 %	-0.6 %	-6.9 %	-59 %
	AU ^d	+4.4 %	+3.5 %	+6.1 %	+67 %
Observations		31,382	13,648	13,696	946
% obs. below QL		0 %	4 %	0 %	7 %
Cluster variation		✓	✓	✓	
Power		✓	✓	✓	✓

Table 3: Effects of animal production on downstream surface water quality in Iowa, 2004–2021 — Difference-in-differences specifications (D4)-(D5). Variables with a *d* superscript are inverse-distance weighted. Standard errors are clustered by drainage area. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimates significant at the 5% level are shown in bold.

Iowa — Precipitation event-study analysis

Spec.	X	DO deficit	TKN _{whole}	TN _{whole}	TN _{diss.}	TP _{whole}	TP _{diss.}
Level effect							
(1)	PE	0.01*** (2.3E-3)	1.4E-3*** (3.3E-4)	2.8E-3 (1.9E-3)	3.3E-3 (3.2E-3)	1.2E-3*** (2.0E-4)	1.9E-4** (8.5E-5)
(2)	PE ^d	0.25** (0.121)	0.053*** (0.016)	0.397*** (0.15)	0.603*** (0.201)	0.037*** (7.0E-3)	9.7E-3*** (3.6E-3)
(3)	PE ^d	0.61** (0.276)	0.072*** (0.021)	0.481** (0.192)	0.09 (0.124)	0.06*** (0.015)	8.6E-3 (0.013)
	PE ^{d2}	-0.037* (0.021)	-2.0E-3 (2.2E-3)	-0.012 (0.042)	0.065*** (0.012)	-2.5E-3 (1.5E-3)	1.4E-4 (1.4E-3)
(4)	PE ^d	0.232* (0.121)	0.053*** (0.015)	0.396*** (0.148)	0.606*** (0.201)	0.037*** (6.9E-3)	9.7E-3*** (3.6E-3)
	PE ^{L1,d}	0.315*** (0.093)	-0.012** (5.2E-3)	0.14** (0.066)	0.078* (0.046)	-5.1E-4 (2.5E-3)	-8.6E-4 (2.7E-3)
Sample mean							
		15.4	0.875	8.47	7.33	0.292	0.157
Relative effect							
(1)	PE	+0.07 %	+0.16 %	+0.03 %	+0.04 %	+0.4 %	+0.12 %
(2)	PE ^d	+1.6 %	+6.1 %	+4.7 %	+8.2 %	+13 %	+6.2 %
(3)	PE ^d +PE ^{d2}	+3.8 %	+8.2 %	+5.6 %	+1.7 %	+20 %	+5.5 %
(4)	PE ^d	+1.5 %	+6.1 %	+4.7 %	+8.3 %	+13 %	+6.2 %
	PE ^{L1,d}	+2 %	-1.34 %	+1.6 %	+1.1 %	-0.18 %	-0.55 %
Observations		31,382	13,648	1,639	1,152	13,696	946
% obs. below QL		0 %	4 %	1 %	1 %	0 %	7 %
Cluster variation		✓	✓		✓	✓	
Power		✓	✓	✓	✓	✓	✓

Table 4: Effects of extreme precipitation events at the location of AFOs on downstream surface water quality in Iowa, 2004–2021. Variables with a *d* superscript are inverse-distance weighted. Standard errors are clustered by drainage area. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimates significant at the 5% level are shown in bold.

North Carolina — Precipitation event-study analysis

Spec.	X	DO deficit	Fecal coliforms	TKN _{whole}	TN _{whole}	TP _{whole}
Level effect						
(1)	PE	0.013*** (3.6E-3)	0.245** (0.12)	1.6E-4*** (5.0E-5)	-1.7E-4 (1.7E-4)	6.6E-5** (3.3E-5)
(2)	PE ^d	0.225*** (0.072)	5.04* (2.94)	3.2E-3* (1.7E-3)	-1.9E-3 (4.5E-3)	2.3E-3** (1.0E-3)
(3)	PE ^d	0.147 (0.128)	13.9** (5.42)	7.2E-3** (2.8E-3)	0.012 (8.8E-3)	3.3E-3*** (1.1E-3)
	PE ^{d2}	3.3E-3 (2.7E-3)	-0.36** (0.154)	-1.6E-4** (6.9E-5)	-8.4E-4* (4.6E-4)	-6.2E-5 (4.6E-5)
(4)	PE ^d	0.216*** (0.07)	4.97* (2.92)	3.0E-3* (1.7E-3)	-1.9E-3 (4.5E-3)	2.3E-3** (9.9E-4)
	PE ^{L1,d}	0.367*** (0.067)	2.95 (2.69)	3.4E-3** (1.5E-3)	-5.5E-3 (4.4E-3)	-5.3E-4 (1.3E-3)
Sample mean						
		15.2	368	0.587	0.966	0.108
Relative effect						
(1)	PE	+ 0.09 %	+ 0.07 %	+ 0.03 %	-0.02 %	+ 0.06 %
(2)	PE ^d	+ 1.5 %	+1.4 %	+0.54 %	-0.2 %	+ 2.1 %
(3)	PE ^d +PE ^{d2}	+0.99 %	+ 3.7 %	+ 1.2 %	+1.2 %	+3.1 %
(4)	PE ^d	+ 1.4 %	+1.4 %	+0.52 %	-0.19 %	+ 2.1 %
	PE ^{L1,d}	+ 2.4 %	+0.8 %	+ 0.57 %	-0.57 %	-0.49 %
Observations		17,416	12,200	10,398	1,143	1,096
% obs. below QL		0 %	0 %	10 %	3 %	0 %
Cluster variation		✓	✓	✓		
Power		✓	✓	✓	✓	✓

Table 5: Effects of extreme precipitation events at the location of AFOs on downstream surface water quality in North Carolina, 2014–2021. Variables with a *d* superscript are inverse-distance weighted. Standard errors are clustered by drainage area. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimates significant at the 5% level are shown in bold.

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Appendix

A.1 Harmonization of the water quality data

The measurements recorded in the Water Quality Portal (WQP) are provided by multiple agencies, and present many inconsistencies in reporting, such as being labeled under different “characteristic names” as well as reported in various units. [Shaughnessy et al. \(2019\)](#) notes that among the most inconsistently reported water quality parameters are some of the most important (and indeed those of interest in this paper): nutrients, including the various nitrogen-containing compounds (e.g., nitrate, nitrite, ammonia) and forms of phosphorus (e.g., orthophosphate, total phosphorus). Indeed, we find in the raw collected data that a unique pollutant, e.g., nitrite, can appear as a concentration *as element* (e.g., “Nitrite-N” in mg/L as N), as a concentration *as polyatomic* (e.g., “Nitrite” in mg NO₂/L), but often also ambiguously as a concentration in mg/L of “Nitrite”, without specifying whether the quantity is in polyatomic or elemental mass.

I harmonize the data by extracting from the WQP all the “characteristic names” that correspond to the parameter of interest, then do extensive data filtering to remove ambiguous or inconsistent measures, and convert all measures to elemental mass concentrations. I average all measurements by station-by-day, to avoid cases of double reporting that would induce the over-representation of specific stations in my final sample. The list of characteristic names considered and the corresponding number of observations collected — pre-data processing — is detailed in Table [A1](#).

Water quality data are also prone to measurement error, which may lead to extreme values beyond the natural value range (e.g., levels of dissolved oxygen exceeding 20 mg/L) or beyond the detection upper bound of the test used. Measurements below the given test’s detection lowerbound, i.e., “non-detects”, are also sometimes erroneously labeled as “0” values. For records on the left tail of the distribution, I follow the approach of the existing literature and transform measures of zero and non-detects as positive values ([Keiser & Shapiro, 2019](#)), specifically to half of the smallest positive value in the sample, following [Raff and Meyer \(2021\)](#). I apply a different correction on the right tail of the distribution, which is generally much larger: I censor data to the maximum physically possible level when one exists (such as for dissolved oxygen), and winsorize at the 99.5% level otherwise.

A.2 Power screening for water-quality outcomes

Water-quality parameters differ sharply in monitoring frequency, sample size, and noise. As a result, some outcome–specification pairs may be incapable of detecting effects of plausible magnitude even if such effects exist. To avoid presenting uninformative null results, I conduct an ex ante power screening that answers: *given the realized panel structure and residual variability, would the regression design have adequate power to detect moderate effects?*

Throughout, power is driven by (i) the number of independent clusters (monitoring stations), (ii) within-cluster variation in the regressor after fixed effects, and (iii) residual variance—rather than by the raw number of observations.

The screening proceeds in three steps.

1. Pre-simulation diagnostics: feasibility checks I first apply fast diagnostics to rule out designs that are mechanically weak under fixed effects and clustered inference:

- Minimum number of clusters. Outcomes with fewer than 30 clusters are excluded from further analysis, as cluster-robust inference becomes unreliable with small numbers of clusters and power is typically low even when the nominal sample size is large.
- Pathological cluster structure. I flag outcomes where identification is likely to collapse after fixed effects:
 - High share of singleton clusters. Those contribute little to identification under clustered inference as they provide no within-cluster contrast once fixed effects are included.
 - Insufficient clusters with within-cluster variation in the regressor. I filter for a minimum number of clusters (20) with at least two observations and nonzero within-cluster standard deviation in the key regressor. This feasibility condition ensures that a substantial subset of clusters contains identifying variation after fixed effects, without requiring every cluster to be informative.

These diagnostics are used as triage tools: they identify cases where power is predictably negligible before running more intensive calculations.

2. Analytical minimum detectable effect (MDE) Next, I compute a theoretical minimum detectable effect (MDE) for each outcome using the baseline clustered fixed-effects specification. The MDE is the smallest effect size that would be detected with 80% power at a 5% two-sided significance level, given the estimated standard error of the treatment coefficient.

Formally, letting G denote the number of clusters and $\widehat{se}_{\hat{\beta}}$ the cluster-robust standard error for the treatment coefficient, the MDE is

$$\text{MDE} = \left(t_{1-\alpha/2, G-1} + t_{1-\gamma, G-1} \right) \widehat{se}_{\hat{\beta}}$$

with $\alpha = 0.05$ and $1 - \gamma = 0.80$.

To express detectability in a scale that matches the identifying variation in the regression, I normalize the MDE by the residual standard deviation from the fitted model, that is, $\text{MDE}_{\text{std}} = \text{MDE}/\text{sd}(\hat{u})$ where $\hat{u}$ are the model residuals. This standardization aligns the MDE with the conditional noise that governs inference under fixed effects and controls. Outcomes with large standardized MDEs (e.g., exceeding 1.5 residual SDs) are treated as weakly informative.

3. Simulation-based power analysis under the realized panel design Finally, I estimate power via simulations that preserves the empirical cluster structure and fixed effects. For each outcome and setting (DiD and event-study), I consider the same treatment, fixed-effects, and control structure as in the baseline regression.

I consider a grid of standardized true effects expressed in units of the residual standard deviation: $e \in 0.25, 0.50, 0.75, 1.00$. Each e corresponds to a change of $e \times \text{sd}(\hat{u})$ in the outcome for a one-standard-deviation increase in the regressor.

Let $\hat{\mu}_i$ denote the fitted value from the baseline model excluding the treatment contribution (i.e., holding the treatment effect at zero), and let $\tilde{x}_i$ be the standardized regressor. For each effect size e , I generate synthetic outcomes

$$y_i^{(s)}(e) = \hat{\mu}_i + e \text{sd}(\hat{u}) \tilde{x}_i + \tilde{u}_i^{(s)}$$

where $\tilde{u}_i^{(s)}$ are simulated errors constructed to respect within-cluster dependence. In practice, I use a wild cluster residual resampling scheme (cluster-level sign flips of residuals) so that the simulated errors retain the empirical heteroskedasticity and cluster correlation patterns.

For each simulated dataset $s = 1, \dots, J$, (with $(J=2000)$), I re-estimate the same clustered fixed-effects regression and record whether the null of no effect is rejected at $\alpha = 0.05$. The estimated power at effect size e is the rejection frequency across simulations.

The simulation yields pairs $(e_k, \hat{p}_k)$ over the effect-size grid. The empirical MDE is then the smallest effect e such that $\hat{p}(e) \geq 0.80$. Because the grid is discrete, I linearly interpolate between the two nearest points bracketing 0.80. If power does not reach 0.80 at the largest tested effect, the empirical MDE is recorded as not attainable on the grid, and I classify the outcome as underpowered.

Finally, as decision rule, I retain outcomes for the main analysis when the simulation indicates adequate power—specifically, when power exceeds 0.8 for moderate standardized effects (approximately 0.3–0.5 residual SDs). Outcomes that fail the feasibility checks, exhibit large analytical MDEs, or display very low simulated power even at large effects are treated as underpowered, so as to not mistake a lack of statistical sensitivity for evidence of no effect.

A.3 Water quality standards and recommended criteria

The U.S. EPA develops drinking water regulations, which impose notably maximum contaminant levels (MCL) for key pollutants of concern. For numerous other compounds present in water, while there is no enforceable national standard, the U.S. EPA provides recommended ambient water quality criteria (AWQC) for the different types of waterbodies, notably rivers and streams. For example, in 2013, the final acute AWQC for protecting freshwater organisms from the toxicity of ammonia was set to 17 mg/L total ammonia nitrogen, and the final chronic AWQC for ammonia to 1.9 mg/L (in the conditions of a pH of 7.0 and a temperature of 20°C) (EPA 822-R-13-001).

For nutrient concentrations, specifically total nitrogen and total phosphorus, recommended criteria are provided for each of the fourteen major “ecoregions” of the country, alongside guidelines for states and authorized tribes for establishing their own water quality standards, in order to protect three designated uses: aquatic life, recreation, and drinking water supply. The criteria proposed differ across ecoregions in order to account for unique local conditions. However, neither North Carolina nor Iowa has developed numeric water

quality criteria for nitrogen and phosphorus.¹¹

Table A2 summarizes the national drinking water standards and recommended surface water quality criteria for the nutrient and pathogen concentrations of concern. Most of Iowa is located in Ecoregion VI (“Corn Belt And Northern Great Plains”), and North Carolina belongs, from East to West, to Ecoregions XIV (“Eastern Coastal Plain”), IX (“Southeastern Temperate Forested Plains and Hills”) and XI (“Central and Eastern Forested Uplands”).

Parameter	Characteristic Name	NC	IA
Dissolved Oxygen (DO)	Dissolved oxygen (DO)	60,633	98,308
	Dissolved oxygen saturation	11,059	6,709
	Oxygen	4,998	9,537
	TOTAL	76,690	114,554
Fecal coliform	Fecal Coliform	22,019	1,803
	TOTAL	22,019	1,803
Total Phosphorus (TP)	Orthophosphate	5,843	97,750
	Orthophosphate as P		11,952
	Phosphate		15,549
	Phosphate-phosphorus	9	36,741
	Phosphate-phosphorus as P		18,349
	Phosphorus	28,772	8,758
	Total Phosphorus, mixed forms	330	2,099
	TOTAL	34,954	191,198
Total Kjeldahl nitrogen (TKN)	Kjeldahl nitrogen	26,591	47,660
	Total Kjeldahl nitrogen		69
	Total Kjeldahl nitrogen (Organic N & NH ₃)	1	80
	TOTAL	26,592	47,809
Total Nitrogen (TN)	Nitrogen	1,980	2,852
	Nitrogen, mixed forms (NH ₃), (NH ₄), or- ganic, (NO ₂) and (NO ₃)	3,944	9,423
	Total Nitrogen, mixed forms	338	773
	TOTAL	6,262	13,048

Notes: CFU = colony-forming units.

Table A1: Characteristic names recorded on the Water Quality Portal, and corresponding number of measurement records by state and period.

¹¹The U.S. EPA publishes the status of states’ numeric nutrient criteria development online: <https://www.epa.gov/nutrient-policy-data/state-progress-toward-developing-numeric-nutrient-water-quality-criteria>

Parameter	Unit	Drinking water standard	Surface water quality criteria
Total Nitrogen	mg/L as N		By Ecoregion: (VI) 2.18, (IX) 0.69, (XI) 0.31, (XIV) 0.71 National recommended limits: 0.1 in flowing waters, 0.05 in streams entering lakes.
Total Phosphorus	mg/L as P		By Ecoregion: (VI) 0.07625, (IX) 0.03656, (XI) 0.010, (XIV) 0.03125
Fecal coliform	cfu/100mL	MCL = 0; Public health goal = 0	

Notes: CFU = colony-forming units; MCL = maximum contaminant level.

Table A2: National drinking water standards and recommended surface water quality criteria for nutrient and pathogen concentrations.

A.4 Robustness checks

Iowa — DiD analysis — Specifications (D4)-(D4alt)					
Spec.	X	DO deficit	TKN _{whole}	TP _{whole}	TP _{diss.}
Level effect					
(4)	AFO $\geq 1,000, d$	4.38***	0.084**	-3.8E-3	0.122**
		(1.02)	(0.037)	(0.026)	(0.058)
	AFO $< 1,000, d$	0.024	9.4E-3	-2.7E-3	-0.011
		(0.382)	(0.016)	(7.7E-3)	(0.015)
(4')	AU $\geq 1,000, d$	0.643**	0.034**	7.7E-3	0.063***
		(0.295)	(0.017)	(0.011)	(0.014)
	AU $< 1,000, d$	1.37**	0.014	-7.4E-3	-0.018
		(0.623)	(0.034)	(0.013)	(0.024)
(4)	AFO $\geq 960, d$	2.84***	0.047*	5.9E-3	0.054
		(0.58)	(0.024)	(0.015)	(0.037)
	AFO $< 960, d$	-0.124	0.014	-7.7E-3	-0.015
		(0.348)	(0.014)	(6.9E-3)	(0.023)
(4')	AU $\geq 960, d$	0.782***	0.029**	8.4E-3	0.045***
		(0.287)	(0.012)	(9.3E-3)	(0.013)
	AU $< 960, d$	1.21	0.02	-0.017	-0.026
		(0.804)	(0.038)	(0.017)	(0.039)
Sample mean					
		15.4	0.875	0.292	0.157
Relative effect					
(4)	AFO $\geq 1,000, d$	+28 %	+9.6 %	-1.3 %	+78 %
	AFO $< 1,000, d$	+0.2 %	+1.1 %	-0.9 %	-6.9 %
(4')	AU $\geq 1,000, d$	+4.2 %	+3.9 %	+2.6 %	+40 %
	AU $< 1,000, d$	+8.9 %	+1.6 %	-2.5 %	-11 %
(4)	AFO $\geq 960, d$	+18 %	+5.3 %	+2 %	+34 %
	AFO $< 960, d$	-0.8 %	+1.6 %	-2.6 %	-9.6 %
(4')	AU $\geq 960, d$	+5.1 %	+3.3 %	+2.9 %	+29 %
	AU $< 960, d$	+7.8 %	+2.3 %	-5.9 %	-16 %
Observations		31,382	13,648	13,696	946
% obs. below QL		0 %	4 %	0 %	7 %
Cluster variation		✓	✓	✓	
Power		✓	✓	✓	✓

Table A3: Effects of animal production on downstream surface water quality in Iowa, 2004–2021 — Difference-in-differences specifications (D4)-(D4alt). Variables with a d superscript are inverse-distance weighted. Standard errors are clustered by drainage area. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimates significant at the 5% level are shown in bold.

Iowa — Precipitation event-study analysis – *Reference-period = week*

Spec.	X	DO deficit	TKN _{whole}	TN _{whole}	TN _{diss.}	TP _{whole}	TP _{diss.}
Level effect							
(1)	PE	0.016*** (2.8E-3)	2.2E-3*** (4.3E-4)	5.9E-4 (1.9E-3)	-9.0E-4 (1.4E-3)	1.5E-3*** (2.9E-4)	2.5E-4 (1.8E-4)
(2)	PE ^d	0.29* (0.174)	0.088*** (0.019)	0.22 (0.197)	0.336** (0.159)	0.049*** (9.7E-3)	0.013 (9.5E-3)
(3)	PE ^d	0.811** (0.36)	0.15*** (0.038)	0.232 (0.223)	-0.437*** (0.133)	0.112*** (0.023)	-1.0E-3 (0.03)
	PE ^{d2}	-0.068** (0.033)	-8.3E-3* (4.6E-3)	-2.4E-3 (0.057)	0.128*** (0.014)	-8.7E-3*** (3.2E-3)	2.5E-3 (4.3E-3)
(4)	PE ^d	0.242 (0.176)	0.089*** (0.019)	0.218 (0.196)	0.352** (0.164)	0.049*** (9.7E-3)	0.013 (9.4E-3)
	PE ^{L1,d}	0.387*** (0.124)	-0.014 (8.7E-3)	0.421*** (0.124)	0.359*** (0.088)	-1.5E-3 (3.9E-3)	-2.5E-3 (4.8E-3)
Sample mean							
		15.4	0.875	8.47	7.33	0.292	0.157
Relative effect							
(1)	PE	+ 0.1 %	+ 0.3 %	+0.01 %	-0.01 %	+ 0.5 %	+0.16 %
(2)	PE ^d	+1.9 %	+ 10 %	+2.6 %	+ 4.6 %	+ 17 %	+8.5 %
(3)	PE ^d +PE ^{d2}	+ 5.2 %	+17 %	+2.7 %	-5.5 %	+ 38 %	-0.2 %
(4)	PE ^d	+1.6 %	+ 10 %	+2.6 %	+ 4.8 %	+ 17 %	+8.6 %
	PE ^{L1,d}	+ 2.5 %	-1.6 %	+ 5 %	+ 4.9 %	-0.5 %	-1.6 %
Observations		31,382	13,648	1,639	1,152	13,696	946
% obs. below QL		0 %	4 %	1 %	1 %	0 %	7 %
Cluster variation		✓	✓		✓	✓	
Power		✓	✓	✓	✓	✓	✓

Table A4: Effects of extreme precipitation **weeks** at the location of AFOs on downstream surface water quality in Iowa, 2004–2021. Variables with a *d* superscript are inverse-distance weighted. Standard errors are clustered by drainage area. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimates significant at the 5% level are shown in bold.

North Carolina — Precipitation event-study analysis – *Reference-period* = *week*

Spec.	X	DO deficit	Fecal coliforms	TKN _{whole}	TN _{whole}	TP _{whole}
Level effect						
(1)	PE	0.017*** (5.3E-3)	0.277** (0.141)	2.2E-4*** (7.5E-5)	-1.9E-4 (2.4E-4)	1.0E-4** (4.9E-5)
(2)	PE ^d	0.373*** (0.104)	3.76 (4.02)	3.0E-3 (2.1E-3)	-4.2E-3 (6.9E-3)	3.0E-3** (1.4E-3)
(3)	PE ^{d2}	-0.013 (8.9E-3)	-0.698* (0.362)	-1.9E-4 (1.6E-4)	-6.4E-3* (3.7E-3)	-2.2E-4 (7.3E-4)
(3)	PE ^d	0.623** (0.246)	17.4* (10.1)	6.9E-3 (4.2E-3)	0.053 (0.035)	5.0E-3 (6.5E-3)
(4)	PE ^d	0.393*** (0.103)	3.83 (4.03)	3.2E-3 (2.1E-3)	-4.5E-3 (6.9E-3)	3.1E-3** (1.5E-3)
(4)	PE ^{d,L1}	0.53*** (0.093)	2.38 (4.36)	4.4E-3** (1.9E-3)	-3.4E-3 (5.2E-3)	6.4E-4 (1.3E-3)
Sample mean						
		15.2	368	0.587	0.966	0.108
Relative effect						
(1)	PE	+ 0.11 %	+ 0.08 %	+ 0.04 %	-0.02 %	+ 0.09 %
(2)	PE ^d	+ 2.5 %	+1 %	+0.51 %	-0.44 %	+ 2.8 %
(3)	PE ^d +PE ^{d2}	+4.1 %	+4.7 %	+1.2 %	+5.3 %	+4.6 %
(4)	PE ^d	+ 2.6 %	+1 %	+0.55 %	-0.47 %	+ 2.9 %
(4)	PE ^{d,L1}	+ 3.5 %	+0.65 %	+ 0.76 %	-0.36 %	+0.59 %
Observations		17,416	12,200	10,398	1,143	1,096
% obs. below QL		0 %	0 %	10 %	3 %	0 %
Cluster variation		✓	✓	✓		
Power		✓	✓	✓	✓	✓

Table A5: Effects of extreme precipitation **weeks** at the location of AFOs on downstream surface water quality in North Carolina, 2014–2021. Variables with a *d* superscript are inverse-distance weighted. Standard errors are clustered by drainage area. Significance codes: *** $p < 0.01$, ** $p < 0.05$, * $p < 0.1$. Estimates significant at the 5% level are shown in bold.